

MATH 341: Homework 3

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Math 341 Numerical Linear Algebra for Deep Neural Networks, Hw 3

(At least one of the problems will be graded)

Exercise 1 Let

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix} \in \mathbb{R}^{3 \times 4}.$$

- (a) Row-reduce A to an echelon form E with one way of tracking row operations by augmenting with I_3 as

$$[A|I_3] \longrightarrow [E|P],$$

so that $PA = E$. Write down E and P , and let r be the number of nonzero rows of E .

$$\begin{array}{ll}
 \xrightarrow{\text{Proposition}} & \left[\begin{array}{cccc|ccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 \\ 2 & 4 & 1 & 3 & 0 & 1 & 0 \\ 1 & 2 & 1 & 2 & 0 & 0 & 1 \end{array} \right] \xrightarrow{R_2=R_2-2R_1} & \left[\begin{array}{cccc|ccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 0 & 0 & 1 \end{array} \right] \\
 \xrightarrow{R_3=R_3-R_1} & \left[\begin{array}{cccc|ccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 0 & 1 \end{array} \right] \xrightarrow{R_3=R_3-R_2} & \left[\begin{array}{cccc|ccc} 1 & 2 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right] \\
 \xrightarrow{R_1=R_1-R_3} & \left[\begin{array}{cccc|ccc} 1 & 2 & 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right] \xrightarrow{R_2=R_2+2R_3} & \left[\begin{array}{cccc|ccc} 1 & 2 & 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & 1 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right]
 \end{array}$$

$$E = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad P = \begin{bmatrix} 0 & 1 & -1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \quad r = 2$$

- (b) Identify the pivot columns of A and give a basis for $R(A)$ using the corresponding columns of the original matrix A . State $\dim(R(A))$.

$$A_{REF} = E = \begin{bmatrix} \mathbf{1} & 2 & 0 & 1 \\ 0 & 0 & \mathbf{1} & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The **pivots** indicate the following pivot columns:

$$\text{Pivot Cols} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\} \quad \text{Basis of } \text{Col}(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$$

(c) Give a basis for $R(A^T)$ using the nonzero rows of E . State $\dim(R(A^T))$.

$$\text{Basis of } Row(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}^T \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}^T \right\}$$

$$\dim(Col(A)) = \dim(Row(A)) = [\dim(A) - \dim(Nul(A))]$$

(d) Solve $Ax = 0$ using E (i.e., solve $Ex = 0$). Give a basis for $N(A)$ and state $\dim(N(A))$.

$$A_{REF} = E = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 + 2x_2 + 0 + x_4 = 0 \\ 0 + 0 + x_3 + x_4 = 0 \end{bmatrix} \Rightarrow \begin{bmatrix} x_1 = -2x_2 + -x_4 \\ x_3 = -x_4 \\ x_2 = x_2 \\ x_4 = x_4 \end{bmatrix}$$

$$x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \Rightarrow Nul(A) = \begin{bmatrix} -2 & -1 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{bmatrix}$$

(e) Find a basis for $N(A^T)$ using P (do not row-reduce A^T):

(i) Let p_i^T be the i -th row of P . Use $PA = E$ to identify $p_i^T A$ for $i = 1, 2, 3$.

$$PA = E \Rightarrow \begin{bmatrix} 0 & 1 & -1 \\ 0 & -1 & 2 \\ 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$p_1^T A = 0^T \Rightarrow \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}$$

$$p_2^T A = 0^T \Rightarrow \begin{bmatrix} 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

$$p_3^T A = 0^T \Rightarrow \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

(ii) For each zero row of E (rows $r + 1, \dots, 3$) use $p_i^T A = 0^T$ and transpose it to obtain $A^T p_i = 0$.

$$p_3^T A = 0^T \Rightarrow \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ 1 & 2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow A^T p_3 = 0 \Rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- (iii) Conclude that the vectors $p_{r+1}, \dots, p_3$ (written as column vectors) form a basis of $N(A^T)$. Write this basis explicitly and state $\dim(N(A^T))$.

$$E_{3,[1,4]} : P_{3,[1,3]} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \implies \text{Basis}(N(A^T)) = \{P_{3,[1,3]}\}$$

$$\dim(\text{Nul}(A)) = \dim(\text{Nul}(A^T)) = 1$$

- (f) Check the dimension identities:

$$\begin{bmatrix} \dim(R(A)) & = & \dim(R(A^T)) & = & r \\ \dim(N(A)) & = & 4 - r, \\ \dim(N(A^T)) & = & 3 - r. \end{bmatrix}$$

$$\text{Recall: } \dim(X) = | \{ \vec{x} : \vec{x} \in X \} |$$

Theorem:

$$\text{Column space: } \dim(\text{Col}(A)) = \dim(\text{Col}(A^T))$$

$$\text{Row space: } \dim(\text{Col}(A^T)) = \dim(\text{Col}(A))$$

$$\text{Rank-nullity: } \dim(A) - \dim(\text{Nul}(A)) = \dim(A) - \dim(\text{Nul}(A^T))$$

$$\text{Theorem: } [\text{Col}(A) = \text{Row}(A^T)] \wedge [\text{Row}(A) = \text{Col}(A^T)]$$

$$\begin{bmatrix} \dim(R(A)) & = & \dim(R(A^T)) & = & r \\ \dim(N(A)) & = & 4 - r, \\ \dim(N(A^T)) & = & 3 - r. \end{bmatrix} \equiv \begin{bmatrix} 2 \\ 4 - 2 = 2 \\ 3 - 2 = 1 \end{bmatrix}$$

- (g) Pick a nonzero $y \in N(A^T)$ from part (e) and verify y is orthogonal to each basis vector of $R(A)$ from part (b) by computing dot products. Pick a nonzero $x \in N(A)$ from part (d) and verify x is orthogonal to each basis row-vector of $R(A^T)$ from part (c).

$$y = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \quad x = \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = 1(1) + (-1)2 + 1(1) = 0$$

$$\begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} = 0(1) + -1(1) + 1(1) = 0$$

$$\begin{bmatrix} -2 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 & 1 \end{bmatrix} = -2 \cdot 1 + 1 \cdot 2 + 0 + 0 = -2 + 2 = 0$$

$$\begin{bmatrix} -2 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 & 1 \end{bmatrix} = 0 + 0 + 0 + 0 = 0$$

- (h) Let $b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$. Use one nonzero $y \in N(A^T)$ to decide whether $b \in R(A)$ by computing $y^T b$ and interpreting the result: is $Ax = b$ consistent?

$$y^T b = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = 1(1) + 0 + 1(1) = 2 \neq 0 \implies \text{Inconsistent}$$

Exercise 2 Let $Q \in \mathbb{R}^{n \times n}$ and let its columns be $q_1, \dots, q_n$.

$$Q = \begin{bmatrix} a_{1,1} & 0_{1,2} & \cdots & 0_{1,n} \\ 0_{2,1} & \ddots & & a_{2,n} \\ \vdots & & \ddots & \\ 0_{n,1} & & & a_{n,n} \end{bmatrix} = [q_1, q_2, \dots, q_n]$$

(a) Compute $(Q^T Q)_{ij}$ in terms of q_i and q_j . Then state what $Q^T Q = I$ says about the columns $q_1, \dots, q_n$.

$$Q^T Q = I \implies (Q^T Q)_{ij} = \begin{cases} i = j & \implies q_i^T q_j = 0 \\ i \neq j & \implies q_i^T q_j = 1 \end{cases}$$

$$(Q^T Q)_{ij} \implies Q \text{ is an orthonormal set}$$

(b) Assume $Q^T Q = I$. Take determinants of both sides and simplify to conclude $\det(Q) \in \{+1, -1\}$.

$$Q^T Q = I \implies \det(Q^T Q) = \det(I)$$

$$I = \begin{bmatrix} 1_{1,1} & 0_{1,2} & \cdots & 0_{1,n} \\ 0_{2,1} & \ddots & & a_{2,n} \\ \vdots & & \ddots & \\ 0_{n,1} & & & 1_{n,n} \end{bmatrix}$$

$$\implies \det(I) = 1_{1,1} \cdot 1_{2,2} \cdot \dots \cdot 1_{n,n} = 1$$

$$\forall A, \det(A) = \det(A^T) \implies \det(Q^T Q) = \det(Q)\det(Q) = \det(Q)^2 = \det(I) = 1$$

$$(\det(Q))^2 = 1 \implies \det(Q) = \pm 1$$

(c) Assume $Q_1^T Q_1 = I$ and $Q_2^T Q_2 = I$. Compute and simplify $(Q_1 Q_2)^T (Q_1 Q_2)$ to verify that $Q_1 Q_2$ is orthogonal.

$$(Q_1 Q_2)^T = Q_2^T Q_1^T$$

$$\implies (Q_2^T Q_1^T)(Q_1 Q_2) = Q_2^T (Q_1^T Q_1) Q_2 = Q_2^T (I) Q_2 = Q_2^T Q_2 = I$$

$$\therefore Q_1 Q_2 \text{ is orthogonal}$$

Exercise 3 (Householder reflector) Let $u \in \mathbb{R}^n$ satisfy $u^T u = 1$ and define

$$H = I - 2uu^T.$$

(a) Compute and simplify $H^T H$ (expand and use $u^T u = 1$) to verify that $H^T H = I$.

$$\text{Note: } (uu^T)^T = (u^T)^T u^T = uu^T$$

$$\begin{aligned} H^T H &= (I - 2uu^T)^T (I - 2uu^T) \\ &= (I - 2uu^T)(I - 2uu^T) \quad \text{Let } 2uu^T = x \\ &= (I - x)(I - x) \\ &= I(I - x) - x(I - x) \\ &= I^2 - Ix - Ix + x^2 \\ &= I - 2Ix + x^2 \\ x^2 &= (2uu^T)^2 = (4uu^T uu^T) \\ &= I - 4Iuu^T + 4uu^T uu^T \\ &= I - 4Iuu^T + 4u(u^T u)u^T \\ &= I - 4Iuu^T + 4uu^T \\ &\therefore H^T H = I \end{aligned}$$

(b) Verify that $H^T = H$ and $H^2 = I$.

$$H^T H = (I - 2uu^T)^T (I - 2uu^T) = HH \implies H^T = H$$

$$\implies H^T H = HH = H^M = I \text{ By pf. above}$$